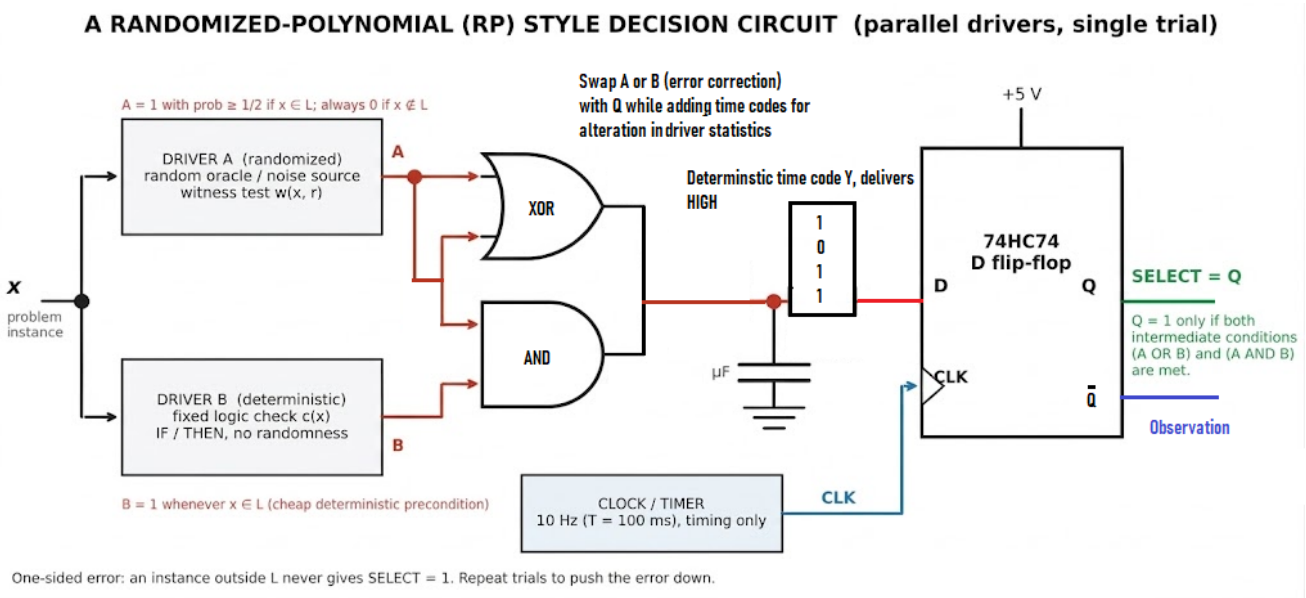
# From Electrons to Photon Statistics

Re-reading the RP-style decision circuit as a model of how electron handling shapes the light an optoelectronic device emits or detects




The source circuit: parallel drivers A and B, XOR/AND gating, time code Y and a 74HC74 flip-flop

**Description of the circuit.** Driver A is a randomized witness test: if x is in L it outputs 1 with probability at least 1/2, and if x is not in L it always outputs 0. Driver B is a deterministic check that outputs 1 whenever x is in L. The two outputs go to an XOR gate and an AND gate, meet at a node with a capacitor, and pass through the fixed time code Y = 1011 to the D input of a 74HC74 flip-flop. A 10 Hz clock tells the flip-flop when to sample, and Q (SELECT) holds the one-bit verdict, with Q-bar as the observation output. Because A is always 0 outside L, SELECT can never be 1 for a NO instance: the error is one-sided, and repeating trials shrinks the remaining error as $(1/2)^k$.

## 1. The claim

In electronics and optoelectronics, **the way electrons are manipulated determines the statistical distribution of the photons emitted or processed**. If electrons arrive at a junction as a random, uncorrelated stream, the photons come out random too. If the electrons are metered out one at a time, the photons inherit that regularity. The RP circuit is a convenient frame for this: it has a *random* branch, a *deterministic* branch, a gating stage, a smoothing capacitor, a timing code and a clocked latch. Each has a physical counterpart in a light-emitting or light-detecting device.

## 2. Three numbers that describe photon statistics

| Quantity | Definition | Reading |
|---|---|---|
| Fano factor F | F = Var(n) / ⟨n⟩ for photons counted in a window | F = 1 Poissonian (laser, random stream). F > 1 super-Poissonian (thermal, bunched). F < 1 sub-Poissonian (regulated, antibunched). |
| Mandel Q | Q = F − 1 | Q = 0, > 0, < 0 for the three cases above. Only Q < 0 has no classical-wave explanation. |
| $g^{(2)}(0)$ | Probability of two photons arriving together, relative to chance | Thermal light: 2. Laser: 1. Ideal single-photon emitter: 0. |

## 3. Re-labelling the circuit

| Circuit element | Electronic / optoelectronic counterpart | Statistical effect |
|---|---|---|
| Input x | The excitation condition: bias, injected carriers, optical pump | Sets the mean photon number. |
| Driver A (random oracle / noise source) | Uncorrelated carrier injection: shot noise, spontaneous recombination, thermal carriers | Pushes the photon count toward Poissonian or, with thermal mixing, super-Poissonian. |
| Driver B (deterministic check) | Regulated injection: constant-current drive through a high impedance, a single-electron turnstile, Coulomb blockade in a quantum dot | Electrons arrive in a regular pattern, so photons are sub-Poissonian. |
| AND gate | Coincidence condition: an electron *and* a hole must both be present before radiative recombination can occur | Photon only if both carriers are available; gates the emission. |
| XOR gate with feedback note (“swap A or B with Q”) | Error-correcting feedback from output to pump, or mismatch detection between random and regulated channels | Feedback that suppresses pump noise is the standard route to amplitude-squeezed light. |
| Capacitor node (µF) | Junction or circuit capacitance with series resistance; charging energy in a single-electron device | Sets the bandwidth over which noise is smoothed or blocked. |
| Time code Y = 1011 | A gated pump-pulse pattern: excite in time bins 1, 3 and 4, skip bin 2 | Fixes *when* photons may appear, shaping the temporal statistics. |
| Clock, 10 Hz | Pulsed excitation or detector gating (real devices: kHz to GHz) | Defines the counting window; 10 Hz is symbolic. |
| 74HC74 D flip-flop, Q = SELECT | Photodetector plus threshold discriminator and latch: a “click” | Converts the photon stream into a recorded 0/1 outcome per window. |
| Q-bar (Observation) | The complementary “no click” record | Same information, opposite polarity. |

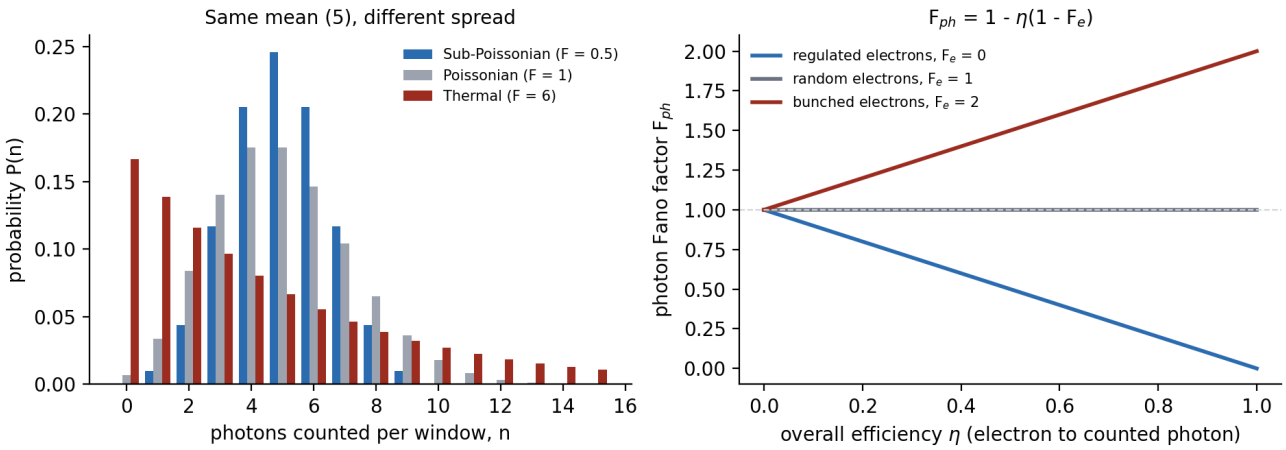
## 4. How electron handling sets the photon distribution

Suppose each injected electron produces a counted photon with overall probability η (internal quantum efficiency times extraction times detector efficiency), independently of the others. Random thinning of a stream gives a simple relation between the electron Fano factor $F_e$ and the photon Fano factor $F_{ph}$:

$$\mathbf{F_{ph} = 1 - \eta\,(1 - F_e)}$$

Three regimes follow directly:

- **Random electrons ($F_e = 1$)**, the Driver A case: $F_{ph} = 1$ whatever η is. The photons are Poissonian.
- **Regulated electrons ($F_e = 0$)**, the Driver B case: $F_{ph} = 1 - \eta$. With $\eta = 0.8$ the photon stream has $F = 0.2$, strongly sub-Poissonian. The regularity survives only as far as the efficiency allows.
- **Bunched electrons ($F_e > 1$)**, for example trap-limited or avalanche-like injection: $F_{ph} > 1$, super-Poissonian.




Left: photon-number distributions with the same mean. Right: the Fano-factor relation above.

## 5. One-sided error becomes detector behaviour

The circuit's guarantee was: *no witness, no accept*. A photon detector behaves the same way. With no photon present there is (apart from dark counts) no click, so false accepts are essentially absent. A photon that is present can still be missed, with probability $1 - \eta$, which plays the role of the false-reject chance ($\leq$ 1/2 in the RP definition, so it needs $\eta \geq 1/2$ to match).

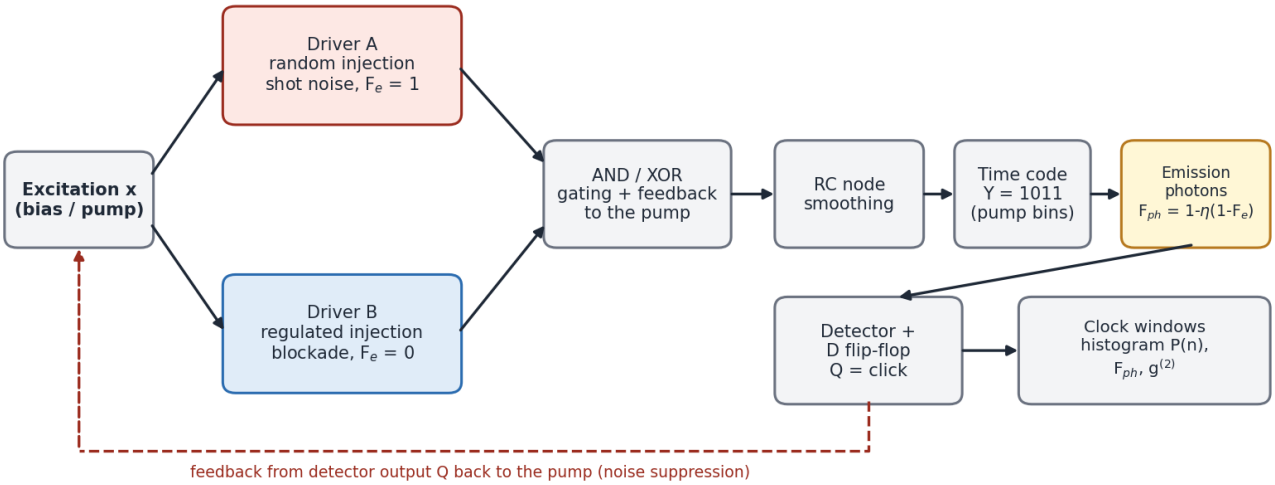
Repeating trials has a direct counterpart too: counting over k clock windows drives the chance of missing a real signal down to $(1 - \eta)^k$. With $\eta = 0.5$, ten windows leave about 0.1%. Dark counts are the one place the analogy breaks cleanly: they are false accepts, which the ideal RP circuit never has, and they accumulate as you add windows.

## 6. Walking through the re-labelled circuit

1. **Excite.** The bias or pump (x) goes to both the random and regulated electron channels.
2. **Inject.** Channel A delivers carriers at random times. Channel B delivers them in a metered, nearly regular pattern.
3. **Gate.** The AND stage lets recombination proceed only when carrier pairs coincide. The feedback path trims pump fluctuations.
4. **Smooth and time.** The RC node filters fast noise and the pulse pattern 1011 fixes the allowed emission bins.
5. **Detect and latch.** On each clock edge the detector stage records click or no click.
6. **Accumulate.** The histogram of clicks over many windows is the photon-number distribution, and its Fano factor reveals which electron statistics were in charge.

## 7. Integration: one chain from electrons to counted photons

Putting the circuit and the physics together gives a single chain. The excitation feeds a random channel and a regulated channel in parallel. Their outputs are gated, smoothed by the RC node and sliced into time bins by the code Y. The resulting electron stream drives emission, and the photons are then thinned by the overall efficiency $\eta$ before the detector and flip-flop turn them into clicks. A feedback path from the output back to the pump closes the loop, which is how pump-noise suppression is achieved in practice.




Integrated block diagram: red path = random branch and feedback, blue = regulated branch

| Operating mode | Which branch dominates | Photon statistics out |
|---|---|---|
| A only | Random injection, no regulation | $F_{ph} = 1$ (Poissonian), or above 1 with thermal mixing. |
| B only | Regulated injection | $F_{ph} = 1 - \eta$ (sub-Poissonian). |
| A and B with AND gating | Photon only when both conditions coincide | Intermediate: noise from A is limited by the regulated gate from B. |
| A and B with feedback via Q | Output corrects the pump | $F_{ph}$ pushed below 1, limited by η and loop bandwidth (set by the RC node). |

## 8. Limits of the analogy

- The circuit is a *decision* device with a single bit out. Photon statistics are a property of many windows, so the mapping is conceptual, not a circuit you could build to emit light.
- The relation $F_{ph} = 1 - \eta(1 - F_e)$ assumes independent loss and one photon per electron. Real devices add correlations from recombination dynamics, cavity effects and detector dead time.
- The drawing's gate wiring (XOR and AND outputs tied together) has logic issues, so only the A AND B reading, which keeps the one-sided guarantee, is used here.
- Strongly sub-Poissonian emission in practice needs low temperature, high efficiency and low noise. Real LEDs and lasers usually sit near F = 1.

**Summary:** random electrons give random photons, regulated electrons give regular photons, and every loss or gate between the two pulls the distribution toward Poisson. The circuit's random driver, deterministic driver, gating, smoothing, timing and latch map onto exactly those steps.